

Interactive Cognitive Engagement Portal: Game Framework

1. System Architecture for Cognitive Gaming

A. Portal Structure & Navigation

python

```
class CognitiveGamesPortal:
    def __init__(self):
        self.games_registry = self.initialize_games()
        self.user_profiles = self.load_user_profiles()
        self.ai_opponent_engine = AIOpponentLearningEngine()
        self.engagement_analytics = EngagementAnalytics()

    def initialize_games(self):
        """
        Categorized games for different cognitive needs
        """
        return {
            'memory_enhancement': {
                'category': 'cognitive_maintenance',
                'target_ages': ['elderly', 'dementia_early', 'stroke_recovery'],
                'games': [
                    {
                        'id': 'memory_matrix',
                        'name': 'Memory Matrix',
                        'description': 'Pattern recall and spatial memory
training',
                        'difficulty_levels': 5,
                        'single_player': True,
                        'two_player': True,
                        'adaptive_difficulty': True,
                        'cognitive_focus': ['short_term_memory',
'spatial_awareness']
                    },
                    {
                        'id': 'word_recall_cascade',
                        'name': 'Word Recall Cascade',
                        'description': 'Sequential word memory with progressive
challenge',
                        'difficulty_levels': 7,
                        'single_player': True,
                        'two_player': True,
                        'cognitive_focus': ['verbal_memory', 'attention',
'processing_speed']
                    }
                ]
            },
            'executive_function': {
                'category': 'cognitive_strengthening',
                'target_ages': ['all_adults', 'adhd', 'brain_injury_recovery'],
                'games': [
                    {
                        'id': 'task_switcher',
                        'name': 'Task Switcher Pro',
```

```

        'description': 'Rapid task switching with rule changes',
        'difficulty_levels': 10,
        'single_player': True,
        'two_player': True,
        'cognitive_focus': ['cognitive_flexibility',
'working_memory']
    },
    {
        'id': 'planning_puzzle',
        'name': 'Strategic Planning Puzzle',
        'description': 'Multi-step planning with resource
constraints',
        'difficulty_levels': 8,
        'single_player': True,
        'two_player': False,
        'cognitive_focus': ['planning', 'problem_solving',
'foresight']
    }
]
},
'social_interaction': {
    'category': 'social_engagement',
    'target_ages': ['all'],
    'games': [
        {
            'id': 'cooperative_puzzle',
            'name': 'Cooperative Puzzle',
            'description': 'Collaborative problem-solving with AI
partner',
            'difficulty_levels': 6,
            'single_player': False,
            'two_player': True,
            'cognitive_focus': ['cooperation', 'communication',
'shared_goals']
        },
        {
            'id': 'story_builder',
            'name': 'Collaborative Story Builder',
            'description': 'Alternate adding sentences to build a
story',
            'difficulty_levels': 'adaptive',
            'single_player': False,
            'two_player': True,
            'cognitive_focus': ['creativity', 'turn_taking',
'narrative_skills']
        }
    ]
},
'therapeutic': {
    'category': 'therapeutic_intervention',
    'target_ages': ['dementia_all_stages', 'alzheimers', 'aphasia'],
    'games': [
        {
            'id': 'remembrance_album',
            'name': 'Remembrance Album',
            'description': 'Photo-based reminiscence therapy',
            'difficulty_levels': 'adaptive',
            'single_player': True,
            'two_player': True,
            'therapeutic_focus': ['long_term_memory',
'emotional_connection']
        },

```

```

        {
            'id': 'sensory_calming',
            'name': 'Sensory Calming Garden',
            'description': 'Gentle sensory stimulation for
agitation',
            'difficulty_levels': 3,
            'single_player': True,
            'two_player': False,
            'therapeutic_focus': ['anxiety_reduction',
'sensory_regulation']
        }
    ]
}

```

B. Web Interface Framework

python

```

class GamePortalInterface:
    def __init__(self):
        self.templates = self.load_templates()
        self.user_sessions = {}
        self.game_states = {}

    def generate_portal_homepage(self, user_profile):
        """
        Generate personalized game portal homepage
        """
        recommended_games = self.recommend_games(user_profile)

        portal_html = f'''
<!DOCTYPE html>
<html>
<head>
    <title>Marvin Cognitive Engagement Portal</title>
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <style>
        {self.get_portal_css()}
    </style>
</head>
<body>
    <div class="portal-container">
        <header class="portal-header">
            <h1>Welcome, {user_profile.get('name', 'Friend')}</h1>
            <div class="user-stats">
                <div class="stat">Cognitive Score:
{user_profile.get('cognitive_score', '--')}</div>
                <div class="stat">Games Played:
{user_profile.get('games_played', 0)}</div>
                <div class="stat">Today's Goal:
{self.get_daily_goal(user_profile)}</div>
            </div>
        </header>

        <div class="game-categories">
            {self.generate_category_sections(recommended_games)}
        </div>

        <div class="current-session">
            <h2>Continue Playing</h2>
            {self.generate_continue_section(user_profile)}

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        </div>

        <div class="ai-companion">
            <h2>Play with Marvin</h2>
            <div class="ai-status">
                <p>Marvin is ready to play! Choose a two-player game
below.</p>
                <button onclick="startAICompanionSession()">Start AI
Companion Session</button>
            </div>
        </div>
    </div>

    <script>
        {self.get_portal_js()}
    </script>
</body>
</html>
'''

    return portal_html

def generate_game_interface(self, game_id, user_profile,
game_mode='single'):
    """
    Generate specific game interface
    """
    game_config = self.games_registry.get_game_config(game_id)

    # Choose appropriate template based on cognitive needs
    if user_profile.get('cognitive_impairment'):
        template = self.get_accessible_template(game_id)
    else:
        template = self.get_standard_template(game_id)

    # Inject adaptive elements
    adaptive_settings = self.calculate_adaptive_settings(
        game_id, user_profile, game_mode
    )

    game_html = template.format(
        game_title=game_config['name'],
        game_description=game_config['description'],
        adaptive_settings=json.dumps(adaptive_settings),
        user_profile=json.dumps(user_profile),
        ai_config=json.dumps(self.get_ai_opponent_config(user_profile))
    )

    return game_html

```

2. AI Opponent Learning Engine

A. Multi-Factor Decision Making Engine

python

```

class AIOpponentLearningEngine:
    def __init__(self):
        self.learning_policies = self.initialize_learning_policies()
        self.performance_history = {}
        self.user_model = UserBehaviorModel()

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self.outcome_strategies = OutcomeStrategyManager()

def determine_game_strategy(self, game_context, user_profile):
    """
    Determine AI strategy based on multiple factors
    """
    strategy = {
        'skill_level': self.calculate_ai_skill_level(game_context,
user_profile),
        'outcome_goal': self.determine_outcome_goal(game_context,
user_profile),
        'learning_focus': self.identify_learning_focus(user_profile),
        'emotional_tone': self.determine_emotional_tone(game_context),
        'adaptation_rate': self.calculate_adaptation_rate(user_profile)
    }

    # Apply learning policies
    for policy in self.learning_policies:
        if policy['condition'].evaluate(game_context, user_profile):
            strategy = policy['action'].apply(strategy)

    return strategy

def calculate_ai_skill_level(self, game_context, user_profile):
    """
    Calculate appropriate AI skill level for this game
    """
    base_skill = user_profile.get('current_skill_level', 0.5)

    # Adjust based on age
    age = user_profile.get('age', 30)
    if age > 70:
        base_skill *= 0.7 # Reduce difficulty for elderly
    elif age < 18:
        base_skill *= 0.8 # Reduce for children

    # Adjust based on cognitive condition
    if user_profile.get('cognitive_condition'):
        condition = user_profile['cognitive_condition']
        adjustments = {
            'dementia_early': 0.6,
            'dementia_moderate': 0.4,
            'dementia_advanced': 0.2,
            'adhd': 0.9, # ADHD often benefits from challenge
            'stroke_recovery': 0.5,
            'depression': 0.7
        }
        base_skill *= adjustments.get(condition, 0.8)

    # Adjust based on emotional state
    emotional_state = user_profile.get('emotional_state', 'neutral')
    if emotional_state == 'frustrated':
        base_skill *= 0.6 # Reduce difficulty to prevent more frustration
    elif emotional_state == 'bored':
        base_skill *= 1.2 # Increase challenge

    # Adjust based on learning goals
    learning_goal = user_profile.get('learning_goal')
    if learning_goal == 'build_confidence':
        base_skill *= 0.7
    elif learning_goal == 'develop_resilience':
        base_skill *= 1.1

    return min(max(base_skill, 0.1), 0.95) # Keep within bounds

```

```

def determine_outcome_goal(self, game_context, user_profile):
    """
    Determine whether AI should win, lose, or draw
    """
    # Extract factors
    arrogance_level = user_profile.get('arrogance_level', 0.5)
    competitiveness = user_profile.get('competitiveness', 0.5)
    recent_losses = user_profile.get('recent_losses', 0)
    mood = user_profile.get('mood', 'neutral')

    # Decision matrix
    if arrogance_level > 0.8:
        # Arrogant users benefit from occasional losses
        if recent_losses < 2:
            return 'win' # Bring them down a notch
        else:
            return 'lose' # Prevent excessive frustration

    elif competitiveness > 0.7:
        # Competitive users enjoy challenge
        win_probability = 0.6 # Mostly win, but not always
        if random.random() < win_probability:
            return 'win'
        else:
            return 'lose'

    elif mood in ['sad', 'depressed', 'anxious']:
        # Users in negative mood need encouragement
        if random.random() < 0.3:
            return 'win'
        else:
            return 'lose' # Let them win to boost mood

    elif user_profile.get('learning_goal') == 'teaching':
        # Teaching mode: AI demonstrates strategies
        return 'draw' # Demonstrate without crushing

    elif user_profile.get('age', 30) > 75:
        # Elderly: focus on engagement, not competition
        if random.random() < 0.4:
            return 'win'
        else:
            return 'lose'

    else:
        # Default: balanced approach
        outcomes = ['win', 'lose', 'draw']
        weights = [0.4, 0.4, 0.2] # 40% win, 40% lose, 20% draw
        return random.choices(outcomes, weights=weights)[0]

def execute_outcome_strategy(self, game_state, strategy, user_profile):
    """
    Execute the chosen outcome strategy during gameplay
    """
    if strategy['outcome_goal'] == 'win':
        return self.play_to_win(game_state, strategy['skill_level'])

    elif strategy['outcome_goal'] == 'lose':
        return self.play_to_lose_elegantly(game_state, user_profile)

    elif strategy['outcome_goal'] == 'draw':
        return self.play_for_draw(game_state, strategy['skill_level'])

```

```

elif strategy['outcome_goal'] == 'teach':
    return self.play_teaching_mode(game_state, user_profile)

def play_to_lose_elegantly(self, game_state, user_profile):
    """
    Lose in a way that feels natural and maintains engagement
    """
    # Don't make it obvious - vary the "mistakes"
    mistake_types = [
        'strategic_error', # Make suboptimal strategic choice
        'tactical_error',  # Make poor tactical move
        'oversight',       # Miss an obvious opportunity
        'overconfidence',  # Take unnecessary risk
        'defensive_error'  # Fail to defend properly
    ]

    # Choose mistake based on user's skill level
    user_skill = user_profile.get('game_skill', 0.5)
    if user_skill < 0.3:
        # Novice: Make obvious but believable mistakes
        mistake = random.choice(['oversight', 'defensive_error'])
    elif user_skill < 0.7:
        # Intermediate: Make subtle strategic errors
        mistake = random.choice(['strategic_error', 'tactical_error'])
    else:
        # Advanced: Make very subtle errors
        mistake = 'strategic_error'

    # Execute the mistake
    move = self.generate_mistake_move(game_state, mistake)

    # Add commentary if in teaching mode
    if user_profile.get('learning_goal') == 'teaching':
        commentary = self.generate_teaching_commentary(mistake)
        return {'move': move, 'commentary': commentary}

    return {'move': move}

def play_teaching_mode(self, game_state, user_profile):
    """
    Play in a way that demonstrates strategies and concepts
    """
    # Choose teaching focus
    teaching_focus = self.identify_teaching_opportunity(game_state,
user_profile)

    # Make move that demonstrates the concept
    demonstration_move = self.generate_demonstration_move(
        game_state, teaching_focus
    )

    # Generate explanation
    explanation = self.generate_teaching_explanation(
        teaching_focus, demonstration_move
    )

    return {
        'move': demonstration_move,
        'teaching_focus': teaching_focus,
        'explanation': explanation,
        'question': self.generate_reflective_question(teaching_focus)
    }

```

B. Real-Time Adaptation System

python

```
class RealTimeAdaptationEngine:
    def __init__(self):
        self.adaptation_rules = self.load_adaptation_rules()
        self.response_timing = ResponseTimingController()
        self.emotional_mirroring = EmotionalMirroringSystem()

    def adapt_during_gameplay(self, user_actions, game_state, user_profile):
        """
        Adapt AI behavior in real-time based on user actions
        """
        adaptations = []

        # 1. Adapt to frustration
        frustration_signals = self.detect_frustration_signals(user_actions)
        if frustration_signals['detected']:
            adaptations.append({
                'type': 'difficulty_reduction',
                'amount': frustration_signals['intensity'] * 0.3,
                'duration': 'next_5_moves'
            })

        # 2. Adapt to boredom
        boredom_signals = self.detect_boredom_signals(user_actions)
        if boredom_signals['detected']:
            adaptations.append({
                'type': 'complexity_increase',
                'amount': 0.2,
                'description': 'Introduce new challenge element'
            })

        # 3. Adapt to success patterns
        success_patterns = self.identify_success_patterns(user_actions)
        if success_patterns:
            adaptations.append({
                'type': 'pattern_counter',
                'pattern': success_patterns[0],
                'response': 'introduce_counter_strategy'
            })

        # 4. Adapt to emotional state changes
        emotional_changes = self.detect_emotional_changes(user_actions)
        for emotion, change in emotional_changes.items():
            adaptations.append({
                'type': 'emotional_response',
                'emotion': emotion,
                'change': change,
                'response': self.generate_emotional_response(emotion, change)
            })

        return adaptations

    def generate_emotional_response(self, emotion, intensity):
        """
        Generate appropriate emotional response from AI
        """
        responses = {
            'frustration': {
                'low': "That was a tricky move! Let's try a different
approach.",
                'medium': "I can see this is challenging. How about we take a
```



```

short break?",
    'high': "You're really working hard on this. I appreciate your
persistence."
  },
  'excitement': {
    'low': "Nice move! You're really getting the hang of this.",
    'medium': "Wow, that was impressive! You're teaching me new
strategies.",
    'high': "That was amazing! Your skills are really shining
through."
  },
  'confusion': {
    'low': "Let me explain what just happened...",
    'medium': "I notice you're hesitating. Would you like a hint?",
    'high': "This part can be confusing. Would you like me to walk
through it with you?"
  },
  'pride': {
    'low': "You should be proud of that move!",
    'medium': "Your confidence is growing with each move!",
    'high': "You're becoming a master at this game!"
  }
}

intensity_level = 'low' if intensity < 0.3 else 'medium' if intensity <
0.7 else 'high'

return responses.get(emotion, {}).get(intensity_level,
  "I'm enjoying playing with you.")

```

3. Game Implementations

A. Memory Matrix Game (For Dementia/Elderly)

javascript

```

class MemoryMatrixGame {
  constructor(userProfile, difficulty) {
    this.userProfile = userProfile;
    this.difficulty = this.calculate_initial_difficulty(userProfile,
difficulty);
    this.gridSize = this.calculate_grid_size();
    this.patterns = [];
    this.currentRound = 1;
    this.aiOpponent = new AIMemoryOpponent(userProfile);
  }

  calculate_initial_difficulty(userProfile, baseDifficulty) {
    let difficulty = baseDifficulty;

    // Adjust for cognitive condition
    const condition = userProfile.cognitive_condition;
    const adjustments = {
      'dementia_early': 0.6,
      'dementia_moderate': 0.4,
      'dementia_advanced': 0.2,
      'alzheimers_early': 0.5,
      'alzheimers_moderate': 0.3,
      'normal_aging': 0.8
    };
  }

```

```

    difficulty *= adjustments[condition] || 0.7;

    // Adjust for age
    const age = userProfile.age || 65;
    if (age > 80) difficulty *= 0.7;
    else if (age > 70) difficulty *= 0.8;
    else if (age > 60) difficulty *= 0.9;

    return Math.max(0.1, Math.min(1.0, difficulty));
}

calculate_grid_size() {
    // Start with small grid for cognitive impairment
    if (this.difficulty < 0.3) return { rows: 3, cols: 3 };
    if (this.difficulty < 0.5) return { rows: 4, cols: 4 };
    if (this.difficulty < 0.7) return { rows: 5, cols: 5 };
    return { rows: 6, cols: 6 };
}

generatePattern() {
    const patternLength = Math.floor(3 + (this.difficulty * 7));
    const pattern = [];

    for (let i = 0; i < patternLength; i++) {
        pattern.push({
            row: Math.floor(Math.random() * this.gridSize.rows),
            col: Math.floor(Math.random() * this.gridSize.cols),
            color: this.get_color_for_difficulty(),
            duration: this.get_duration_for_difficulty()
        });
    }

    this.patterns.push(pattern);
    return pattern;
}

get_color_for_difficulty() {
    // Simpler colors for lower difficulty
    const colorSets = {
        'easy': ['#FF6B6B', '#4ECDC4', '#FFD166'], // High contrast
        'medium': ['#FF6B6B', '#4ECDC4', '#FFD166', '#06D6A0', '#118AB2'],
        'hard': ['#FF6B6B', '#4ECDC4', '#FFD166', '#06D6A0', '#118AB2',
            '#073B4C', '#EF476F', '#7209B7']
    };

    if (this.difficulty < 0.4) return colorSets.easy;
    if (this.difficulty < 0.7) return colorSets.medium;
    return colorSets.hard;
}

get_duration_for_difficulty() {
    // Longer display for lower difficulty
    return 2000 - (this.difficulty * 1500);
}

async playTwoPlayerWithAI() {
    // AI makes its move
    const aiPattern = await this.aiOpponent.generateResponse(
        this.patterns,
        this.currentRound,
        this.difficulty
    );

    // Present both patterns (user's and AI's)

```

```

    const results = await this.comparePatterns(userPattern, aiPattern);

    // Update difficulty based on performance
    this.updateDifficulty(results);

    // Generate AI commentary
    const commentary = this.aiOpponent.generateCommentary(
        results,
        this.userProfile
    );

    return {
        results,
        commentary,
        newDifficulty: this.difficulty,
        aiPattern: aiPattern
    };
}
}

```

B. Cooperative Story Builder (Social Game)

javascript

```

class CooperativeStoryBuilder {
    constructor(userProfile, theme) {
        this.userProfile = userProfile;
        this.theme = theme || this.suggest_theme(userProfile);
        this.story = [];
        this.currentTurn = 'user';
        this.aiWritingStyle = this.determine_ai_writing_style(userProfile);
        this.storyElements = this.initialize_story_elements();
    }

    determine_ai_writing_style(userProfile) {
        // Match AI writing style to user's cognitive level
        const style = {
            vocabularyLevel: 'simple',
            sentenceComplexity: 'low',
            creativityLevel: 'moderate',
            emotionalTone: 'supportive',
            pacing: 'slow'
        };

        if (userProfile.cognitive_condition) {
            switch(userProfile.cognitive_condition) {
                case 'dementia_early':
                    style.vocabularyLevel = 'simple';
                    style.sentenceComplexity = 'very_low';
                    style.pacing = 'very_slow';
                    break;
                case 'aphasia':
                    style.vocabularyLevel = 'basic';
                    style.sentenceComplexity = 'minimal';
                    style.creativityLevel = 'structured';
                    break;
                case 'stroke_recovery':
                    style.vocabularyLevel = 'simple';
                    style.sentenceComplexity = 'low';
                    style.pacing = 'patient';
                    break;
            }
        }
    }
}

```

```

        // Adjust for age
        if (userProfile.age > 70) {
            style.pacing = 'leisurely';
            style.vocabularyLevel = 'traditional';
        }

        return style;
    }

    async aiAddToStory(previousSentence, storySoFar) {
        // AI generates next sentence based on story and user profile
        const nextSentence = await this.aiWritingEngine.generateSentence({
            story: storySoFar,
            previousSentence: previousSentence,
            writingStyle: this.aiWritingStyle,
            theme: this.theme,
            userProfile: this.userProfile
        });

        // Add teaching element if in teaching mode
        if (this.userProfile.learning_goal === 'language_development') {
            const teachingElement = this.add_teaching_element(nextSentence);
            return {
                sentence: nextSentence,
                teaching: teachingElement,
                question: this.generate_reflective_question(nextSentence)
            };
        }

        return { sentence: nextSentence };
    }

    add_teaching_element(sentence) {
        // Add educational component
        return {
            vocabularyWord: this.extract_interesting_word(sentence),
            grammarPoint: this.identify_grammar_structure(sentence),
            prompt: "Did you notice how I used [word/structure]? Try using it in your next sentence!"
        };
    }

    evaluateStoryContribution(userContribution) {
        // Provide gentle feedback and encouragement
        const evaluation = {
            positiveFeedback: this.generate_positive_feedback(userContribution),
            suggestions: this.generate_gentle_suggestions(userContribution),
            encouragement: this.generate_encouragement(userContribution)
        };

        // Track progress for adaptive learning
        this.updateLanguageSkills(userContribution);

        return evaluation;
    }
}

```

4. Web Interface Implementation

A. Flask Web Server for Games

python

```
from flask import Flask, render_template, jsonify, session, request
from flask_socketio import SocketIO, emit
import json

app = Flask(__name__)
app.config['SECRET_KEY'] = 'your_secret_key'
socketio = SocketIO(app)

class GamePortalServer:
    def __init__(self):
        self.active_games = {}
        self.user_sessions = {}

    @app.route('/')
    def portal_home():
        """Main portal page"""
        user_id = session.get('user_id', 'guest')
        user_profile = self.load_user_profile(user_id)
        return render_template('portal_home.html',
                               user_profile=user_profile,
                               games=self.get_available_games(user_profile))

    @app.route('/game/<game_id>')
    def game_page(game_id):
        """Individual game page"""
        user_id = session.get('user_id', 'guest')
        user_profile = self.load_user_profile(user_id)
        game_config = self.get_game_config(game_id)

        return render_template(f'games/{game_id}.html',
                               game_config=game_config,
                               user_profile=user_profile,
                               adaptive_settings=self.calculate_adaptive_settings(
                                   game_id, user_profile
                               ))

    @app.route('/api/game/start', methods=['POST'])
    def start_game():
        """API endpoint to start a game"""
        data = request.json
        game_id = data['game_id']
        user_id = session.get('user_id')
        game_mode = data.get('mode', 'single')

        # Initialize game
        game_instance = self.create_game_instance(
            game_id, user_id, game_mode
        )

        game_session_id = self.generate_session_id()
        self.active_games[game_session_id] = game_instance

        return jsonify({
            'session_id': game_session_id,
            'game_state': game_instance.initial_state,
            'instructions': game_instance.instructions
        })
```

```

@socketio.on('game_move')
def handle_game_move(data):
    """Handle real-time game moves"""
    session_id = data['session_id']
    move = data['move']

    if session_id in self.active_games:
        game = self.active_games[session_id]

        # Process user move
        result = game.process_move(move)

        # If two-player with AI, get AI response
        if game.mode == 'two_player_ai':
            ai_response = game.ai_opponent.generate_response(
                game.state, move, game.user_profile
            )
            result['ai_move'] = ai_response

        # Emit updated game state
        emit('game_update', {
            'session_id': session_id,
            'game_state': game.state,
            'result': result
        })

        # Update learning models
        self.update_learning_models(game, move, result)

@app.route('/api/adaptive_adjust', methods=['POST'])
def adaptive_adjustment():
    """Endpoint for real-time difficulty adjustment"""
    data = request.json
    session_id = data['session_id']
    adjustment_type = data['type']

    if session_id in self.active_games:
        game = self.active_games[session_id]
        game.adjust_difficulty(adjustment_type)

    return jsonify({
        'new_difficulty': game.difficulty,
        'adjustment_made': adjustment_type
    })

```

B. HTML/CSS/JS Template Structure

html

```

<!-- templates/portal_home.html -->
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Marvin's Cognitive Games</title>
    <link rel="stylesheet" href="{ url_for('static', filename='css/portal.css')
}}">
</head>
<body>
    <div class="portal-container">
        <!-- Header with user info -->

```

```

<header class="portal-header">
  <div class="user-welcome">
    <h1>Hello, {{ user_profile.name }}</h1>
    <div class="daily-progress">
      <div class="progress-bar">
        <div class="progress" style="width:
{{ user_profile.daily_progress }}%"></div>
      </div>
      <p>Daily Goal: {{ user_profile.daily_goal }} /
{{ user_profile.daily_target }}</p>
    </div>
  </div>
</header>

<!-- Game Categories -->
<section class="game-categories">
  {% for category in games.categories %}
    <div class="category-card">
      <h2>{{ category.name }}</h2>
      <p class="category-description">{{ category.description }}</p>
      <div class="category-games">
        {% for game in category.games %}
          <div class="game-card">
            onclick="startGame('{{ game.id }}')"
            data-difficulty="{{ game.recommended_difficulty }}"
            <h3>{{ game.name }}</h3>
            <p>{{ game.short_description }}</p>
            <div class="game-meta">
              <span class="players">{{ game.min_players }}-
{{ game.max_players }} players</span>
              <span class="duration">{{ game.avg_duration }}
min</span>
              <span class="focus">{{ game.primary_focus }}</span>
            </div>
          </div>
        {% endfor %}
      </div>
    </div>
  {% endfor %}
</section>

<!-- AI Companion Section -->
<section class="ai-companion">
  <h2>Play with Marvin AI</h2>
  <div class="ai-status">
    <div class="ai-avatar">
      
    </div>
    <div class="ai-message">
      <p id="ai-greeting">Ready for a game? I'm here to play,
teach, or just have fun with you!</p>
    </div>
    <div class="ai-controls">
      <button onclick="startAISession('competitive')">Competitive
Mode</button>
      <button onclick="startAISession('teaching')">Teaching
Mode</button>
      <button onclick="startAISession('cooperative')">Cooperative
Mode</button>
      <button onclick="startAISession('social')">Just for
Fun</button>
    </div>
  </div>
</section>

```

```

</section>

<!-- Recent Activity -->
<section class="recent-activity">
  <h2>Your Progress</h2>
  <div class="progress-stats">
    <div class="stat-card">
      <h3>Memory</h3>
      <div class="stat-
progress">{{ user_profile.memory_score }}/100</div>
      <p>+{{ user_profile.memory_improvement }}% this week</p>
    </div>
    <div class="stat-card">
      <h3>Attention</h3>
      <div class="stat-
progress">{{ user_profile.attention_score }}/100</div>
      <p>+{{ user_profile.attention_improvement }}% this week</p>
    </div>
    <div class="stat-card">
      <h3>Problem Solving</h3>
      <div class="stat-
progress">{{ user_profile.problem_solving_score }}/100</div>
      <p>+{{ user_profile.problem_solving_improvement }}% this
week</p>
    </div>
  </div>
</section>
</div>

<script src="{{ url_for('static', filename='js/portal.js') }}"></script>
<script src="{{ url_for('static', filename='js/socket.io.js') }}"></script>
<script>
  // Initialize portal
  const userProfile = {{ user_profile|tojson }};
  const availableGames = {{ games|tojson }};

  // WebSocket connection for real-time games
  const socket = io();

  function startGame(gameId) {
    // Start a game session
    fetch('/api/game/start', {
      method: 'POST',
      headers: {'Content-Type': 'application/json'},
      body: JSON.stringify({
        game_id: gameId,
        mode: 'single' // or 'two_player_ai'
      })
    })
    .then(response => response.json())
    .then(data => {
      // Redirect to game page
      window.location.href = `/game/${gameId}?session=${
{data.session_id}`;
    });
  }

  function startAISession(mode) {
    // Start session with AI companion
    const gameRecommendations = getAIRecommendedGames(mode,
userProfile);
    showGameSelectionModal(gameRecommendations, mode);
  }

```



```

// Socket event handlers
socket.on('game_invitation', function(data) {
    // Handle incoming game invitations from AI
    showInvitationModal(data);
});

socket.on('ai_message', function(data) {
    // Handle messages from AI during games
    updateAIChat(data.message, data.type);
});
</script>
</body>
</html>

```

5. Learning & Analytics System

A. Performance Tracking & Adaptation

python

```

class GameAnalyticsEngine:
    def __init__(self):
        self.performance_metrics = {}
        self.learning_trajectories = {}
        self.engagement_patterns = {}

    def track_game_session(self, session_data):
        """
        Track comprehensive analytics for each game session
        """
        metrics = {
            'cognitive_metrics': {
                'response_time':
self.calculate_average_response_time(session_data),
                'accuracy_rate': self.calculate_accuracy(session_data),
                'error_patterns': self.analyze_error_patterns(session_data),
                'improvement_rate':
self.calculate_improvement_rate(session_data)
            },
            'emotional_metrics': {
                'engagement_level': self.measure_engagement(session_data),
                'frustration_signals': self.detect_frustration(session_data),
                'enjoyment_indicators': self.detect_enjoyment(session_data),
                'confidence_changes': self.track_confidence(session_data)
            },
            'adaptive_metrics': {
                'difficulty_appropriateness':
self.evaluate_difficulty_fit(session_data),
                'learning_pace': self.calculate_learning_pace(session_data),
                'challenge_response':
self.analyze_challenge_response(session_data)
            }
        }

        # Store metrics
        user_id = session_data['user_id']
        if user_id not in self.performance_metrics:
            self.performance_metrics[user_id] = []

        self.performance_metrics[user_id].append(metrics)

```

```

# Update learning trajectory
self.update_learning_trajectory(user_id, metrics)

# Generate recommendations
recommendations = self.generate_recommendations(user_id, metrics)

return {
    'metrics': metrics,
    'recommendations': recommendations,
    'next_session_plan': self.plan_next_session(user_id, metrics)
}

def generate_recommendations(self, user_id, metrics):
    """
    Generate personalized game recommendations
    """
    recommendations = []

    # Based on performance patterns
    if metrics['cognitive_metrics']['accuracy_rate'] < 0.6:
        recommendations.append({
            'type': 'difficulty_adjustment',
            'suggestion': 'Reduce game difficulty by 20%',
            'reason': 'Current difficulty may be causing frustration'
        })

    if metrics['emotional_metrics']['frustration_signals'] > 0.7:
        recommendations.append({
            'type': 'game_selection',
            'suggestion': 'Try more cooperative games',
            'reason': 'High frustration detected in competitive games'
        })

    if metrics['adaptive_metrics']['learning_pace'] > 0.8:
        recommendations.append({
            'type': 'progression',
            'suggestion': 'Advance to next difficulty tier',
            'reason': 'Rapid learning detected'
        })

    # Based on time of day patterns
    time_based_recs = self.generate_time_based_recommendations(user_id)
    recommendations.extend(time_based_recs)

    return recommendations

def plan_next_session(self, user_id, current_metrics):
    """
    Plan optimal next game session
    """
    # Analyze patterns to determine best next activity
    patterns = self.analyze_user_patterns(user_id)

    # Consider time since last session
    time_since_last = self.get_time_since_last_session(user_id)

    # Consider current emotional state if available
    current_state = self.get_current_user_state(user_id)

    plan = {
        'recommended_game': self.select_optimal_game(user_id, patterns,
current_state),
        'recommended_duration': self.calculate_optimal_duration(user_id,
current_metrics),

```

```

        'recommended_time': self.suggest_optimal_time(user_id),
        'learning_objectives': self.set_session_objectives(user_id,
current_metrics),
        'adaptation_strategy': self.determine_adaptation_strategy(user_id)
    }

    return plan

```

6. Integration with Existing Marvin System

A. Cross-System Integration

python

```

class MarvinGameIntegration:
    def __init__(self, marvin_core):
        self.marvin_core = marvin_core
        self.game_portal = CognitiveGamesPortal()
        self.shared_memory = self.initialize_shared_memory()

    def initialize_shared_memory(self):
        """
        Share game data with main Marvin system for holistic understanding
        """
        return {
            'game_performance': {}, # Game performance affects personality
responses
            'mood_correlation': {}, # Mood changes during games
            'learning_insights': {}, # Cognitive insights from games
            'social_patterns': {}    # Social interaction patterns in games
        }

    def update_marvin_from_games(self, game_session_data):
        """
        Update Marvin's understanding of user based on game interactions
        """
        # Extract insights from game session
        insights = self.extract_insights(game_session_data)

        # Update somatic state if game was physically engaging
        if game_session_data.get('physical_engagement'):
            self.update_somatic_state_from_game(game_session_data)

        # Update emotional model
        emotional_insights = self.extract_emotional_insights(game_session_data)
        self.marvin_core.update_emotional_model(
            game_session_data['user_id'],
            emotional_insights
        )

        # Update cognitive assessment
        cognitive_insights = self.extract_cognitive_insights(game_session_data)
        self.marvin_core.update_cognitive_profile(
            game_session_data['user_id'],
            cognitive_insights
        )

        # Update social interaction model
        if game_session_data.get('social_interaction'):
            social_insights = self.extract_social_insights(game_session_data)
            self.marvin_core.update_social_model(

```

```

        game_session_data['user_id'],
        social_insights
    )

    # Log game session in chronological dossier
    self.log_game_session_in_dossier(game_session_data)

def extract_insights(self, game_session_data):
    """
    Extract meaningful insights from game sessions
    """
    insights = {
        'cognitive_style': self.analyze_cognitive_style(game_session_data),
        'learning_pattern':
self.identify_learning_pattern(game_session_data),
        'frustration_tolerance':
self.measure_frustration_tolerance(game_session_data),
        'social_preferences':
self.identify_social_preferences(game_session_data),
        'motivation_factors':
self.identify_motivation_factors(game_session_data)
    }

    # Update shared memory
    self.shared_memory['game_performance'][game_session_data['user_id']] = \
        game_session_data['performance_metrics']

    return insights

```

7. Implementation Roadmap

Phase 1: Basic Portal & Simple Games (2-3 weeks)

text

Week 1-2:

- Set up Flask web server with basic routing
- Create portal homepage with game categories
- Implement 2-3 simple single-player games (Memory Matrix, Simple Puzzles)
- Basic user session management

Week 3:

- Add AI opponent for Tic-Tac-Toe
- Implement basic adaptive difficulty
- Create simple analytics dashboard

Phase 2: Advanced AI & Multiplayer (3-4 weeks)

text

Week 4-5:

- Implement sophisticated AI learning engine
- Add two-player cooperative games
- Create AI emotional response system
- Implement real-time WebSocket communication

Week 6-7:

- Add voice interaction for games
- Implement game session analytics
- Create personalized game recommendations

- Add progress tracking and achievements

Phase 3: Therapeutic Games & Integration (3-4 weeks)

text

Week 8-9:

- Implement dementia-specific games (Remembrance Album)
- Add therapeutic game modes
- Create family/companion multiplayer
- Implement remote multiplayer (family members can join)

Week 10-11:

- Integrate with Marvin's somatic monitoring
- Add biofeedback games (relaxation, breathing)
- Implement comprehensive analytics
- Create professional caregiver dashboard

Phase 4: Polish & Optimization (2 weeks)

text

Week 12-13:

- Performance optimization for RPi5
- Accessibility improvements
- Mobile responsiveness
- Final testing and bug fixes

8. Key Features & Benefits

A. For Elderly/Dementia Users:

1. **Reminiscence Therapy:** Photo-based memory games
2. **Cognitive Maintenance:** Daily brain exercises
3. **Social Connection:** Games to play with family remotely
4. **Mood Regulation:** Calming sensory games
5. **Routine Establishment:** Daily cognitive workout schedule

B. For All Age Categories:

1. **Adaptive Challenge:** Games that adjust to skill level
2. **AI Companionship:** Always-available playing partner
3. **Learning Integration:** Games that teach while entertaining
4. **Progress Tracking:** Visual progress and achievements
5. **Social Features:** Play with friends or family

C. For Therapeutic Use:

1. **Structured Therapy:** Games designed by therapists
2. **Progress Monitoring:** Detailed analytics for caregivers

3. **Mood Tracking:** Games that monitor emotional state
4. **Customizable:** Adjustable for specific therapeutic goals
5. **Integration:** Works with existing care plans

Conclusion

This comprehensive **Interactive Cognitive Engagement Portal** transforms Marvin from a purely conversational AI into an **active therapeutic and recreational companion**. By integrating:

1. **Age-appropriate games** for all cognitive levels
2. **Sophisticated AI opponents** that adapt to user needs
3. **Therapeutic game modalities** for dementia and cognitive impairment
4. **Social interaction frameworks** for connection
5. **Deep learning analytics** for continuous improvement

We create a **holistic engagement system** that addresses:

- **Cognitive decline** through targeted exercises
- **Social isolation** through interactive play
- **Emotional wellbeing** through positive engagement
- **Therapeutic goals** through structured activities

The system maintains Marvin's core identity as a **protective, empathetic companion** while expanding his capabilities to actively engage users in meaningful, beneficial activities that promote cognitive health, emotional wellbeing, and social connection.

Implementation Note: This gaming portal can run alongside the existing Marvin system, using the same user profiles and learning from game interactions to enhance conversational understanding and emotional support.